

Information loss of fixed linear health indicators as a control problem: simplex geometry and reachability limits

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Abstract

A health indicator compresses several condition-monitoring channels into one scalar, and normally keeps the same weights for the life of the component. We take it to be a fixed linear map of the noise-whitened channels. Because degradation runs through several mechanisms at differing rates, the channel combination carrying most information about remaining life rotates as the component ages, so a fixed indicator necessarily discards information. We show that the operating point sets how much it discards, which makes that loss a control variable. The loss reduces to a single angle, between the indicator and the direction along which information currently arrives, and writing each channel's share of that information as a probability vector places that angle on a statistical simplex. Zero loss is attainable exactly when the directions reachable at each instant share a common point, which for generic plants needs one fewer independently adjustable operating variable than there are mechanisms. Scaling every channel by the same factor leaves the loss unchanged, so no criterion built from the Fisher information about remaining life can detect it. On a representative three-mechanism synthetic plant, steering the operating point reduces the information-loss objective defined here by a factor of 68 against the best constant operating point, and a dimensionless index, available before any indicator is chosen, indicates whether such a gain is plausible. Six public run-to-failure fleets

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test the premises separately: the informative direction does rotate on measured cells, and simulated turbofan records confirm against a negative control that the operating point moves it. The same fleets bound the theory: the highest-fidelity turbofan data reject the separable plant form under which the operator’s authority becomes computable offline, so that branch describes a model class rather than measured machinery.

Keywords: prognostics, health indicator, remaining useful life, information loss, optimal control, reachability

1. Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record; the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [4, 2]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

The rotation is not a property of the instrumentation. The degradation physics produces it, and that physics responds to stress: temperature, load and duty cycle change the relative rates of the competing mechanisms, not only the overall speed. Running a cell hotter does not only age it faster;

27 it also shifts the balance between the mechanisms, and therefore moves the
28 direction in which information about remaining life arrives.

29 The operating point is, in most applications, something the operator
30 chooses. So the misalignment between a fixed indicator and the informa-
31 tive direction is not fixed by nature. It is a function of the input, and the
32 operator can steer it.

33 That reframing turns a data-analysis question into a control question, and
34 control questions come with sharper answers than design heuristics do. One
35 can ask what misalignments are reachable at a given operating envelope,
36 whether zero loss is attainable at all, how many independent inputs that
37 would take, and what the best achievable loss is when it is not. These are
38 the questions this paper answers.

39 They sit against five bodies of work. Four address parts of the problem
40 and the fifth supplies the geometry used in Section 3; to our knowledge none
41 of them addresses the position taken here.

42 The prognostics literature treats the compression as a design problem to
43 be solved from data: searching genetically over candidate features against
44 monotonicity and trendability criteria [2], learning it with a recurrent net-
45 work [5], or surveying the resulting pipelines [1, 6]. Two limits on that
46 programme are known, information-theoretic bounds on what any fixed com-
47 pression can retain [3] and the number of run-to-failure units needed to learn
48 one reliably [4]. The first is our own, and the boundary is worth drawing
49 explicitly: it takes one degradation trajectory as given and asks how much
50 information a fixed compression must lose along it. The question here is the
51 complementary one, whether that trajectory can itself be steered so that the
52 same fixed map loses less. Both prior limits hold the trajectory fixed; neither
53 asks whether it can be changed.

54 The predictors downstream are overwhelmingly stochastic — Wiener,
55 gamma and inverse-Gaussian degradation processes with first-passage fail-
56 ure [7, 8], reviewed for remaining-life estimation in [9, 10] and embedded in
57 system-level design methodology in [11]. That literature models how damage
58 evolves, not how the measurement should be compressed before the model is
59 applied, which is the question upstream of it.

60 A separate literature does treat the operating point as a decision variable,
61 but to extend remaining life rather than to preserve estimability [12, 13];
62 the two aims can conflict, and Section 6 exhibits an instance in which they
63 disagree. Choosing inputs to make a state easier to estimate is itself classical,
64 going back to dual control [14], and its modern forms shape a scalarisation

Table 1: Where this paper sits. The columns are the axes that separate these literatures, not a ranking: each body of work answers a question the others do not ask, and the last column is a restriction this paper imposes on itself rather than an advantage over the others. “Input alters degradation” means the operating point enters the degradation dynamics; “direction” means the informative direction of Section 2 is itself an objective.

	input alters degradation	lifetime is an objective	direction is an objective	indicator fixed, scalar
Indicator design [2, 5, 1]	—	—	—	✓
Fixed-indicator limits [3, 4]	—	—	—	✓
Stochastic life models [7, 9]	—	—	—	—
Health-aware control [12, 13]	✓	✓	—	—
Observability-aware [14, 15, 17]	✓	—	partly	—
This paper	✓	secondary	✓	✓

of the Fisher information matrix [15, 16, 17]. Our Proposition 4.6 shows the quantity at issue here to be invisible to every such scalarisation.

The fifth body is information geometry. The reformulation of Section 3 rests on the Fisher–Rao geometry of the probability simplex and on the Bhattacharyya coefficient [18, 19], and the flow it produces is the replicator equation of evolutionary dynamics [20, 21]. These are standard objects; what is new here is their identification with the informative direction of a degrading component, and the fact that the fitness driving the flow is actuated.

The contributions are as follows.

1. The information a fixed indicator discards is an exact function of one misalignment angle (Lemma 2.2). Placing the channels’ shares of the Fisher information on a statistical simplex turns the degradation into a replicator flow whose rates the operating point modulates, and the optimal indicator into a barycentre of the degradation path (Section 3).
2. Zero loss is attainable exactly when the reachable direction sets share a point, which generically needs at least as many independent operating variables as mechanisms less one (Section 4). Supplying the missing input collapses the peak reachability floor from 4.52° to zero at every state, so the rule prices instrumentation rather than only bounding it. Taking logarithms makes every reachable set a translate of one fixed set, which turns the question into a containment and shows the counting rule to be sufficient but not necessary; the exceptions exist for any number of mechanisms and are too fragile to build on (Section 4.1). The count separates two objectives that are easily conflated: holding

- 89 the misalignment at a constant level is a single scalar condition and
90 generically needs one input, whereas freezing the informative direction
91 outright is $d - 1$ conditions. That gap is the difference between useful
92 steering and lossless steering, and an installation with one controllable
93 variable can have the first without the second.
- 94 3. The loss is invariant to a common gain on all channels and is therefore
95 invisible to every scalarisation of the Fisher information about remain-
96 ing life, so no criterion built from that information alone can rank
97 indicators by the loss they incur.
 - 98 4. The best achievable loss is the value of an optimal control problem
99 whose objective is a ratio, and whose terminal time is itself controlled
100 (Section 5), and a dimensionless regime index, computable before any
101 indicator is designed, indicates the regime in which steering can have
102 substantial value, with its accuracy and the direction of its errors mea-
103 sured on 299 randomly drawn plants (Section 6.7).
 - 104 5. The premises are separately testable on public run-to-failure data, and
105 Section 7 tests three in turn: whether the informative direction ro-
106 tates, whether the operating point moves it, and whether the separable
107 stress model survives. The first holds on measured fleets, the second
108 so far only in simulation, and the third is rejected. That section there-
109 fore reports which hypotheses withstand contact with data rather than
110 demonstrating that the method works.

111 The results do not all rest on the same assumptions, and it is worth
112 separating them before any of them is stated. Lemma 2.2, Proposition 2.3
113 and Theorem 4.1 need only the observation model of Section 2: a fixed linear
114 indicator on whitened channels, and a scalar latent. Lemma 4.3 and Propo-
115 sition 4.4 need more, namely that the input enters each mechanism through
116 a single separable factor, and Section 7 reports that the highest-fidelity tur-
117 bofan data reject exactly that. The empirical readings of Section 7 need
118 more again, since estimating one informative direction from a fleet presumes
119 the units differ by a time shift rather than by their degradation coefficients.
120 Fig. 6 is the map, and the reader who wants to know which branch a given
121 result sits on should consult it rather than infer it.

122 Section 2 states the model. Sections 3 to 5 develop the theory, Section 6
123 exercises the three of its claims that are stated sharply enough to be refuted,
124 and Section 7 reports what public fleet data can and cannot establish about
125 the premises the theory rests on.

126 2. Problem formulation

127 This section states the degradation model, says precisely what a prognos-
 128 tic algorithm is being asked to estimate and where the uncertainty in that
 129 quantity comes from, derives the direction along which information about
 130 it arrives, and defines the fixed indicator whose information loss the rest of
 131 the paper studies. Readers familiar with [3] will recognise the model; the
 132 derivation is given in full here because the geometry of Section 3 depends on
 133 several of its details.

134 2.1. Degradation driven by a controlled operating point

135 A component degrades through d distinct mechanisms acting at once. In
 136 a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-
 137 electrolyte interphase and mechanical fatigue of the electrode; in a rolling
 138 bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the
 139 damage accumulated by mechanism i up to time t , and collect these into the
 140 state $x = (x_1, \dots, x_d)$.

141 One externally chosen operating point drives all d mechanisms. It collects
 142 the temperature, the mechanical or electrical load, and the duty cycle at
 143 which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

144 with $\mathcal{U}$ compact, representing the operating envelope the application permits.
 145 Throughout this paper u is the control input: it is the decision variable,
 146 and the question we ask is what choosing it well can buy. The number of
 147 independent inputs, m , turns out to decide what is achievable, and it is
 148 usually much smaller than the number of mechanisms d .

149 The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

150 in which the two factors play different roles:

- 151 • $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much
 152 faster that mechanism runs at operating point u than at a reference
 153 point. It carries all the dependence on the input.
- 154 • $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends
 155 on the damage it has already done. It carries all the dependence on
 156 the state.

157 This separation is the standard accelerated-life form and covers the stress
158 models in routine use [22]. In the Arrhenius model the input is absolute
159 temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential
160 constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's
161 constant. In the inverse-power-law model the input is a mechanical or electri-
162 cal stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress
163 exponent. Choosing h_i recovers the exponential, power-law and stretched-
164 exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and
165 $h_i > 0$.

166 The within-unit dynamics are deterministic, where most of the degrada-
167 tion literature places the randomness inside a single unit's path [7, 8, 10].
168 We put it between units instead (Section 2.2). The object of study is the
169 direction along which information arrives, which is fixed by the relative rates
170 of the competing mechanisms: a within-unit diffusion would jitter that di-
171 rection without altering its systematic motion, whereas dispersion in the rate
172 parameters across a population is what makes remaining life uncertain and
173 therefore worth estimating.

174 We measure damage on a scale normalised so that mechanism i exhausts
175 the component's tolerance when x_i reaches unity. A component is withdrawn
176 at a condemnation limit at or below that point, so we write the end of life
177 as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

178 No result below depends on the value of x_f , which enters only by fixing where
179 a trajectory stops. The computed quantities of Section 6 do depend on it,
180 and the value used there is stated with the rest of the test plant.

181 2.2. What is to be estimated, and why it is uncertain

182 The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

183 the time left before failure.

184 We must say where the uncertainty in R lies, because (2) is a deterministic
185 differential equation: for one component whose initial damage and parame-
186 ters were known exactly, T_f and hence R would be known exactly, and there
187 would be nothing to estimate. The uncertainty is between components, not
188 within one. Units built to the same specification differ in initial damage and

189 in the coefficients of (2), which is the unit-to-unit variability that reliability
 190 engineering treats as the primary source of scatter in life data [22]. A prog-
 191 nostic algorithm observes one unit drawn from that population and must
 192 infer its remaining life.

193 We therefore treat R , at any given observation instant, as a random
 194 variable over the population of units, with prior variance $\text{Var}[R] > 0$. This
 195 variance is the uncertainty the algorithm exists to reduce, and it is the reason
 196 the estimation problem is not trivial. Everything below is written for a
 197 nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

198 2.3. Observations and the informative direction

199 Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

200 with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal
 201 variables. The record available to the algorithm is the collection of these
 202 measurements over the observation window.

203 Equation (5) pairs each mechanism with a channel of its own. Condition
 204 monitoring in general returns $y = H(x, u) + \epsilon$, with sensors that mix mech-
 205 anisms, and the pairing is an assumption rather than a description. It is
 206 load-bearing: the positivity of $\mathbf{d}$ that Remark 3.1 shows the results to need
 207 follows from $\varphi_i > 0$ and $h_i > 0$ in these coordinates and need not survive
 208 an arbitrary H . Where H is known, invertible and positivity-preserving the
 209 construction carries over to $H^{-1}y$; the general case does not reduce to it.

210 Two assumptions carry the sensitivity, and both are stated here rather
 211 than left to be inferred.

212 *The prior on R .* We take R to be Gaussian over the population with variance
 213 $\text{Var}[R]$. Linearising the record about the nominal trajectory, following [3],
 214 then makes the record and R jointly Gaussian, and the mutual informations
 215 below are exact. Linearisation alone would not do it: with a non-Gaussian
 216 prior of the same variance the Gaussian value is an upper bound on what
 217 a channel carries, and Lemma 2.2 becomes an inequality rather than an
 218 identity.

219 *How units differ.* We take the population spread to be a reparametrisation
 220 of a single degradation path: units differ in where they sit on the trajectory
 221 of (2), not in its shape. This is what makes R a coordinate along that path,

222 and it is what the next display uses. Dispersion in the coefficients of (2) is a
 223 different matter, because it moves $\partial x/\partial \theta$ rather than sliding along $\partial x/\partial R$,
 224 and Remark 2.1 records what it costs.

225 The sensitivity of channel i to remaining life then follows from (4): be-
 226 cause R falls at unit rate as t advances,

$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

227 **Remark 2.1** (what the time-shift assumption costs). *Equation (6) is exact*
 228 *when the population spread slides along one path, and only then. On the*
 229 *plant of Section 6.1, with units instead drawn by dispersing the degradation*
 230 *coefficients, the direction that the population regression of x on R actually*
 231 *returns stands about 45° from $-\varphi(u)h(x)$, across two families of dispersed*
 232 *coefficient and every spread tried; a control whose units are started on the*
 233 *reference trajectory, where (6) holds by construction, returns the same re-*
 234 *gression to within 0.2° . The misalignments this paper is about are a few*
 235 *degrees. Extending the construction to populations dispersed in θ , where the*
 236 *observation carries scatter that no value of R explains, is open.*

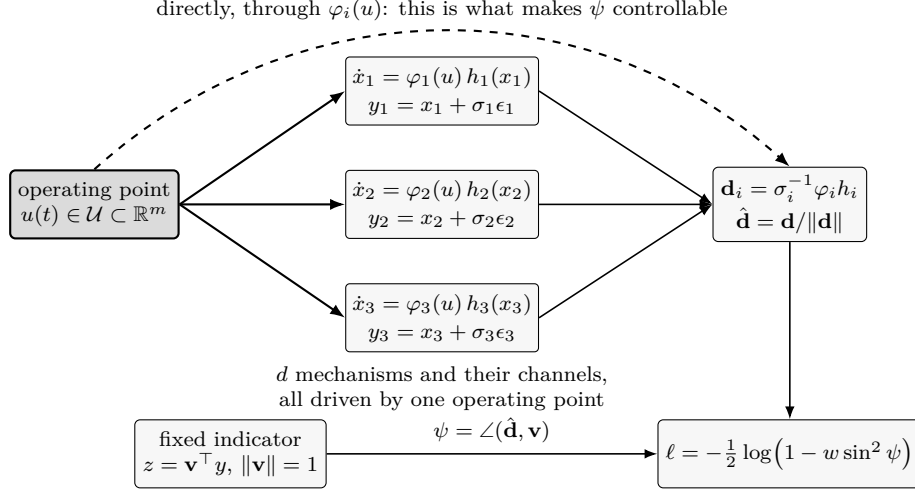
237 Dividing each channel by its own noise level puts them on a common
 238 scale, and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = \left[\sigma_i^{-1} \varphi_i(u) h_i(x_i) \right]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|. \quad (7)$$

239 In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the
 240 measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the
 241 *informative direction*, the combination a health indicator has to be aligned
 242 with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the
 243 strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much
 244 information arrives and $\hat{\mathbf{d}}$ is where it arrives from, and the loss of a fixed
 245 indicator depends only on the second. Experiment design in the observability
 246 tradition scores the first, which is why Proposition 4.6 finds it blind to the
 247 loss.

248 We use two properties of (7) repeatedly below.

249 *Positivity.* Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i >$
 250 0 in (2). This is a consequence of the degradation model rather than an
 251 assumption imposed for convenience, and Remark 3.1 shows that the results
 252 below would fail without it.



Alignment asks $\hat{\mathbf{d}}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

253 *Actuation.* $\mathbf{d}$ is an explicit algebraic function of the state and the input.
 254 The direction along which information about R arrives is therefore not a
 255 fixed property of the instrumentation: it moves when the operating point
 256 moves. That is the observation this paper is built on, and Fig. 1 places it:
 257 the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.
 258 Because R is scalar, a single signal-to-noise ratio governs the information
 259 the record carries about it, namely the squared sensitivity times the prior
 260 uncertainty,

$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

261 Rates in (7) are taken with respect to clock time, whereas [3] differentiates
 262 with respect to age normalised by T_f . The two conventions differ by a factor
 263 common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same
 264 geometry in Sections 3–4, while rescaling γ . Section 5, where lifetime enters
 265 the objective, restores the normalised convention.

266 2.4. The fixed health indicator and its information loss

267 A health indicator compresses the d measured channels into one scalar
 268 before any life prediction is made. We take it to be a fixed linear map.

269 Since (7) already measures each channel in units of its own noise, we write
 270 the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and
 271 $\tilde{y} := \Sigma^{-1/2}y$ the whitened record,

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$

272 with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole
 273 life of the component. Fixing it is not a modelling convenience: the map
 274 is typically embedded in firmware or frozen by certification, so it cannot be
 275 re-tuned as the component ages. An indicator specified on the raw measure-
 276 ments, $z = \mathbf{w}^\top y$, is the same map under $\mathbf{v} \propto \Sigma^{1/2}\mathbf{w}$, and the normalisation
 277 $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters
 278 because the angle below is measured after whitening: unless the σ_i are equal,
 279 a unit vector in raw coordinates is not a unit vector in whitened ones, and
 280 the two readings of ψ disagree.

281 We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle
 282 between the informative direction $\hat{\mathbf{d}}(t)$ and the indicator weights $\mathbf{v}$. Af-
 283 ter whitening, the signal in the record points along $\hat{\mathbf{d}}$ and has magnitude
 284 $\|\mathbf{d}\|\sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit
 285 vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise,
 286 so the indicator's signal-to-noise ratio is

$$\gamma \cos^2 \psi \quad (10)$$

287 against the γ of (8) available to an oracle that is free to re-aim its projection at
 288 $\hat{\mathbf{d}}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio
 289 s carries $\frac{1}{2} \log(1 + s)$ nats about R [23], the information the fixed indicator
 290 forfeits, at each instant, is

$$\ell = \frac{1}{2} \left[\log(1 + \gamma) - \log(1 + \gamma \cos^2 \psi) \right] \geq 0, \quad (11)$$

291 The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

292 a loss per unit time. We optimise this rate rather than the information the
 293 whole observation record forfeits, because (12) is unaffected by how densely
 294 the record is sampled and is therefore a property of the indicator rather

295 than of the measurement schedule. The two are different quantities, and
 296 Proposition 2.3 gives the second in closed form.

297 A third quantity is the one an operator actually pays for: the error of
 298 whatever remaining-life estimator runs downstream. It is neither of the first
 299 two and is not a function of them. The loss is a logarithmic information
 300 quantity and the error is a time, and an estimator holding a whole history
 301 recovers much of what any single instant discards, so a large spread in (12)
 302 does not imply a comparable spread in error. What the theory can offer
 303 downstream is therefore an ordering rather than a conversion, and Section 6.8
 304 tests that ordering against an estimator built from none of this construction:
 305 it survives, with a spread in error much smaller than the spread in loss.
 306 Equation (12) should be read accordingly, as an information-geometric ob-
 307 jective whose ranking carries downstream, not as a surrogate for prognostic
 308 accuracy.

309 Two remarks on what (11) does and does not say. It is a loss relative to
 310 an oracle that no fixed map can realise, so it measures the price of the time-
 311 invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still
 312 pays it whenever $\hat{\mathbf{d}}$ moves. And it depends on the indicator only through the
 313 single angle ψ : two indicators that make the same angle with $\hat{\mathbf{d}}$ are equally
 314 good, whatever else distinguishes them.

315 The next lemma replaces (11) by a closed form that will be more conve-
 316 nient throughout, and records how it behaves at small misalignment.

317 **Lemma 2.2.** *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

318 *exactly and at every signal-to-noise ratio, and*

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

319 *Proof.* Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log(1 - w \sin^2 \psi).$$

320 For (14), use $-\frac{1}{2} \log(1-s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 -$
 321 $\psi^4/3 + O(\psi^6)$; collecting the ψ^4 terms gives $-w/6 + w^2/4 = w(3w-2)/12$. $\square$

322 The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is
 323 therefore the leading term of an exact identity rather than an approximation
 324 of unknown quality. The quartic coefficient $w(3w - 2)/12$ changes sign at
 325 $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic
 326 model overstates the loss, above it the model understates it. This matters in
 327 practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life
 328 measurement with a weak signal on one side, a well-instrumented component
 329 near failure on the other. Fig. 2(a) plots the exact loss against its quadratic
 330 approximation on both sides of that ratio.

331 **Proposition 2.3.** *Let the record consist of measurements (5) at instants*
 332 *$t_1, \dots, t_N$, with the noise independent across channels and across instants,*
 333 *and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1+G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

334 *Then the information about R that the fixed indicator forfeits over the whole*
 335 *record is*

$$\ell_{\text{rec}} = -\frac{1}{2} \log(1 - WS). \quad (15)$$

336 *Proof.* In whitened coordinates the linearised record is $\tilde{y}_k = \mathbf{d}_k R + \epsilon_k$ with
 337 $\epsilon_k \sim N(0, I_d)$ independent, so the Fisher information it carries about R is
 338 $\sum_k \mathbf{d}_k^\top \mathbf{d}_k = \sum_k \|\mathbf{d}_k\|^2$ and an oracle reading the whole record obtains $\frac{1}{2} \log(1 +$
 339 $G)$ nats. The fixed indicator returns the scalars $z_k = \mathbf{v}^\top \tilde{y}_k = (\mathbf{v}^\top \mathbf{d}_k)R + \mathbf{v}^\top \epsilon_k$,
 340 whose noise terms $\mathbf{v}^\top \epsilon_k$ are standard normal and independent because $\|\mathbf{v}\| =$
 341 1 , so the compressed record carries Fisher information $\sum_k (\mathbf{v}^\top \mathbf{d}_k)^2$. Since
 342 $\mathbf{v}^\top \mathbf{d}_k = \|\mathbf{d}_k\| \cos \psi_k$,

$$\text{Var}[R] \sum_k (\mathbf{v}^\top \mathbf{d}_k)^2 = \text{Var}[R] \sum_k \|\mathbf{d}_k\|^2 \cos^2 \psi_k = G(1 - S),$$

343 and subtracting the two mutual informations gives $\frac{1}{2} \log(1 + G) - \frac{1}{2} \log(1 +$
 344 $G(1 - S)) = -\frac{1}{2} \log \frac{1+G-GS}{1+G} = -\frac{1}{2} \log(1 - WS)$. $\square$

345 Equation (15) is (13) again, with the instantaneous signal-to-noise ratio
 346 γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted

by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore survives the passage from an instant to a record. What does not survive is the averaging: $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ. They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises $\mathbf{v}^\top (\sum_k \mathbf{d}_k \mathbf{d}_k^\top) \mathbf{v}$, so the best fixed indicator for the record is the leading eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$.

Corollary 2.4 (what the loss costs an estimator). *In the model of Section 2, the minimum-variance estimate of R formed from the fixed indicator carries mean-squared error*

$$\frac{\text{Var}[R \mid z]}{\text{Var}[R \mid \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}$$

times that of the estimate formed from the whole whitened record, at each instant. The same identity holds over a record with ℓ replaced by ℓ_{rec} of Proposition 2.3.

Proof. Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$ and R Gaussian. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$ and $\text{Var}[R \mid z] = \text{Var}[R]/(1 + \gamma \cos^2 \psi)$. The record's sufficient statistic is $\mathbf{d}^\top \tilde{y}$, with ratio γ , giving $\text{Var}[R \mid \tilde{y}] = \text{Var}[R]/(1 + \gamma)$. Their quotient is $e^{2\ell}$ by (11). The record statement is the same computation with G and S in place of γ and $\sin^2 \psi$. $\square$

The loss is therefore not a score standing in for estimation error: it is the logarithm of the factor by which that error inflates. What the corollary does not settle is whether the ordering survives an estimator other than the one the model prescribes, and Section 6.8 checks that separately.

3. The information simplex and its replicator flow

Lemma 2.2 reduced the loss to one angle. This section shows that the angle lives on a familiar object, the probability simplex, and that the degradation traces a flow along it that the operating point drives. The payoff is threefold: the loss becomes a recognised statistical distance, the optimal indicator becomes a barycentre, and the effect of the input becomes a single interpretable quantity. Two of the results here are load-bearing for what

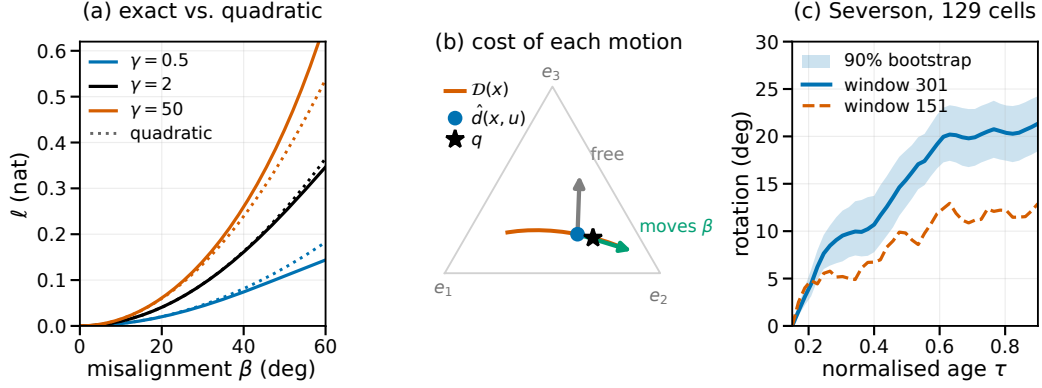


Figure 2: (a) Lemma 2.2: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

377 follows — the replicator form of Theorem 3.4 and the misalignment rate of
 378 Theorem 3.6, which Section 4 builds the reachability argument on. The Bhat-
 379 tacharyya identity and the barycentre are interpretation: they say what the
 380 loss and the optimal indicator already are in recognised terms, and nothing
 381 later depends on them.

382 3.1. Information shares

383 Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components
 384 of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

385 The interpretation is direct. The total Fisher information the record carries
 386 about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i
 387 is exactly the share of the total information about remaining life that arrives
 388 through channel i . A component in which one mechanism dominates the
 389 degradation has p near a vertex of the simplex; one in which three mecha-
 390 nisms contribute equally sits near its centre.

391 The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an
 392 optimal indicator may be taken with non-negative weights.

393 **Remark 3.1** (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$
 394 at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative*

weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$ inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a third of the time. The results therefore require the positive-orthant structure of the degradation model; it is not only a notational convenience.

3.2. The loss as a Bhattacharyya distance

The identification below is used twice in what follows: it gives “the average of a curve on the simplex” an unambiguous meaning (Section 3.3), and it lets the high-signal limit of the loss be read off as a known divergence.

Proposition 3.2. $\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya divergence,

$$\ell = -\frac{1}{2} \log \left[(1 - w) + w e^{-2D_{\text{B}}(p, q)} \right]. \quad (17)$$

As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.

Proof. By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$ is the classical isometry between the simplex under the Fisher–Rao metric and the positive orthant of the unit sphere, under which the geodesic distance is twice the spherical angle [18, 19]; hence $\psi = \frac{1}{2} d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$, and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

The square root is a bijection between the simplex and the non-negative part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition it is an isometry. We use the two descriptions interchangeably from here on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable direction, and a distance between points of the simplex is the angle between the corresponding unit vectors.

The resulting picture is the one the rest of the paper develops. The degrading component is a curve on the simplex, the health indicator is a single point on it, and the information lost depends only on the distance between them. Design of the indicator is the choice of one point; control of the operating point is the shaping of the curve.

426 3.3. The optimal indicator as a barycentre

427 If the indicator is one point and the plant is a curve, the best point should
 428 be some kind of average of the curve. It is, but of two different kinds in two
 429 different limits.

430 **Proposition 3.3.** *Let q^* minimise the time-averaged loss.*

- 431 (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_B$ pointwise, so q^* tends to the Bhattacharyya*
 432 *barycentre $\arg \min_q \int D_B(p_\tau, q) d\tau$.*
 433 (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so q^* tends to the w -weighted*
 434 *Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}\bar{w} \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} :=$*
 435 *$\min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

436 *Proof.* (i) As $\gamma \rightarrow \infty$ we have $w \rightarrow 1$, and (13) gives

$$\ell \rightarrow -\frac{1}{2} \log \cos^2 \psi = -\log \sum_i \sqrt{p_i q_i} = D_B$$

437 by Proposition 3.2. The convergence is uniform, which is what carries the
 438 minimisers with it. Positivity keeps p_τ interior, and $\sqrt{q_i} \geq q_i$ with $\sum_i q_i = 1$
 439 gives $\cos \psi = \sum_i \sqrt{p_i q_i} \geq \min_i \sqrt{p_i}$ for every q in the simplex, so $\cos \psi$ is
 440 bounded away from zero uniformly in q and in τ and D_B stays finite. Both
 441 objectives are continuous on a compact set and uniformly close on it, so
 442 their minimisers converge. (ii) By (14), $\bar{\ell} = \frac{1}{2}\bar{w} \mathbb{E}_\mu[\psi^2] + O(\psi^4)$ with $d\mu =$
 443 $w d\tau/\bar{w}$; minimising a μ -weighted mean square of a geodesic distance is the
 444 definition of the weighted Fréchet mean, and $\psi = \frac{1}{2}d_{\text{FR}}$ converts $\frac{1}{2}\bar{w} \mathbb{E}_\mu[\psi^2]$
 445 into $\frac{1}{8}\bar{w} \mathbb{E}_\mu[d_{\text{FR}}^2]$. $\square$

446 The two limits are taken in different parameters, signal-to-noise ratio in
 447 (i) and arc length in (ii), and they do not coincide. On a curve of angular
 448 span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–
 449 Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5}
 450 degrees of the latter by 50 dB. For design purposes the practical consequence
 451 is mild: either barycentre of a nominal degradation path is a good starting
 452 indicator.

453 3.4. The replicator flow and the role of the input

454 We now ask how fast the informative direction moves and what the op-
 455 erating point does to that motion.

456 **Theorem 3.4.** *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i .*
 457 *Along (2),*

$$\dot{\hat{\mathbf{d}}}_i = \hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

458 *so the information shares obey the replicator equation of evolutionary dynam-*
 459 *ics, in which each share grows in proportion to the amount by which its own*
 460 *rate exceeds the average, the average being taken over the shares themselves:*

$$\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

461 *and*

$$\|\dot{\hat{\mathbf{d}}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = \left[\varphi_i(u) h'_i(x_i) \right]_i + \nabla_u \log \varphi \cdot \dot{u}. \quad (20)$$

462 *Proof.* From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\hat{\mathbf{d}}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has
 463 components $\hat{\mathbf{d}}_i \dot{g}_i - \hat{\mathbf{d}}_i \sum_j \hat{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives
 464 $\text{Var}_p[\dot{g}]$. Since $p_i = \hat{\mathbf{d}}_i^2$, $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\hat{\mathbf{d}}}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i =$
 465 $\log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i)\varphi_i h_i$,
 466 and $(h'_i/h_i)\varphi_i h_i = \varphi_i h'_i$. $\square$

467 Equation (20) has a compact reading. The speed at which the informative
 468 direction rotates equals the information-weighted spread of the mechanisms'
 469 logarithmic rates. If all mechanisms accelerate at the same proportional rate
 470 the direction does not move at all, however fast the component is degrading;
 471 it is disagreement among the mechanisms that rotates it. The identity is the
 472 Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's
 473 fundamental theorem of natural selection [18, 20, 21]. What is new here is
 474 its identification with the informative direction of a degrading component,
 475 and, decisively, that the quantity playing the role of fitness, the logarithmic
 476 rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's
 477 lever.

478 **Remark 3.5** (contrast with Oja's flow). *Oja's flow for online principal-*
 479 *component extraction has the same algebraic form; for a diagonal covariance*
 480 *it reduces to a replicator equation with fitness twice the eigenvalues [24]. The*
 481 *roles are opposite. There the flow is an algorithm, run to fixation in order*
 482 *to find a direction, with constant fitness. Here it is the plant, its fitness is*
 483 *time-varying and actuated, and the aim is to prevent fixation from carrying*
 484 *the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\hat{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.2 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{e}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\hat{\mathbf{d}}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$. Also $\hat{e} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}}) / \sin \beta$ and $\dot{\hat{\mathbf{d}}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle = \dot{\hat{\mathbf{d}}}^\top \mathbf{v} / \sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\dot{\hat{\mathbf{d}}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits below hold for every measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits.

The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d-2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

4. Reachability limits on the misalignment

Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The distinction is the subject of this section, and it is what decides whether a fixed indicator can be made lossless at all.

For a state x , the set of informative directions the operator can produce by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

the reachable direction set, and the closest the indicator can be brought at that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

515 For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures
 516 the actuation authority available at x , and it shrinks to a point when the
 517 mechanisms respond identically to the input.

518 **Theorem 4.1.** *Fix an admissible input and let $x(\cdot)$ be the trajectory it gener-*
 519 *ates. Zero loss along that trajectory is attainable with the fixed indicator q if*
 520 *and only if $q \in \cap_t \mathcal{D}(x_t)$. Solving $\hat{\mathbf{d}}(x, u) = q$ for u imposes $d - 1$ independent*
 521 *scalar equations on m unknowns, so for generic data a common point exists*
 522 *only if $m \geq d - 1$. If $m < d - 1$, then for generic data $\beta_{\min} > 0$ along*
 523 *the trajectory; Remark 4.5 exhibits plants for which it fails. In every case*
 524 *$\bar{\ell} \geq \int \ell(\beta_{\min}(x_t, q)) dt / T_f$ for that trajectory.*

525 *Proof.* By Lemma 2.2, $\ell(\beta) = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\beta_t = 0$ for almost
 526 every t , that is iff $\hat{\mathbf{d}}(x_t, u_t) = q$, that is iff $q \in \mathcal{D}(x_t)$ for every t . Solving
 527 $\hat{\mathbf{d}}(x, u) = q$ imposes $d - 1$ independent scalar equations, both sides being
 528 unit vectors, on m unknowns. If $m < d - 1$ then $\mathcal{D}(x)$ is the image of an
 529 m -dimensional set, hence of dimension at most m , and is a null subset of the
 530 $(d - 1)$ -dimensional sphere that a generic q misses; if $m \geq d - 1$ the implicit
 531 function theorem supplies a solution wherever the Jacobian has full rank.
 532 The inequality $\beta_t \geq \beta_{\min}(x_t, q)$ holds for every admissible input by definition
 533 of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and is strict exactly where $q \notin \mathcal{D}(x_t)$, which
 534 for $m < d - 1$ is generic. Since ℓ is increasing on $[0, \pi/2]$, the bound follows
 535 on dividing by T_f . $\square$

536 This is a state-conditioned result. The bound is genuine, in the sense
 537 that no input violates a reachability constraint at the state it occupies, but
 538 it is conditional on the trajectory, which the input also steers. Optimising
 539 over the trajectory as well is deferred to Section 5, and the two levels are the
 540 structure of the argument: a state-conditioned limit here, a trajectory-level
 541 limit there.

542 The practical reading of Theorem 4.1 is a counting rule: a component
 543 degrading through d mechanisms generically needs $d - 1$ independently ad-
 544 justable operating variables before a fixed indicator can be kept perfectly
 545 aligned. Section 4.1 says what the qualifier covers. Three mechanisms and a
 546 single temperature setpoint will not do it. The three results of this section
 547 answer different questions, and it is worth separating them. Theorem 4.1
 548 characterises zero loss exactly, as membership of every reachable set, and
 549 then counts dimensions to say when a generic plant can meet it; the count is
 550 necessary for generic data and is not by itself a construction. Proposition 4.4

551 gives an exact criterion, necessary and sufficient, on the separable subclass
 552 alone. Remark 4.5 exhibits plants outside that subclass which attain zero
 553 loss with fewer inputs than the count asks, so the count is not necessary in
 554 general. But a weaker and much cheaper objective is available.

555 **Corollary 4.2** (what actuation is actually required). *Two objectives that are*
 556 *easily conflated carry different requirements.*

- 557 (i) *Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d - 1$*
 558 *conditions on $\dot{u}$, hence generically $m \geq d - 1$.*
- 559 (ii) *Holding β at a constant level, not necessarily zero, is the single scalar*
 560 *equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$*
 561 *stays in $\mathcal{U}$; generically $m \geq 1$ suffices.*

562 *Proof.* (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for every t , so $\hat{\mathbf{d}}$ is constant;
 563 by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d - 1$ independent
 564 conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap
 565 between them is the content of that theorem: $\dot{\beta}$ sees only the component of
 566 $\hat{\mathbf{d}}$ along $\hat{e}$, so the remaining $d - 2$ directions of rotation are unconstrained by
 567 (ii) and constrained by (i). $\square$

568 Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the
 569 trajectory that carries it, and is the problem of Section 5.

570 4.1. The reachable sets are translates of one set

571 Theorem 4.1 counts equations against unknowns and hedges with
 572 “generic”. The hedge is not idle, and the following observation says exactly
 573 what it covers.

574 **Lemma 4.3** (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo*
 575 *the all-ones vector, which is what $\hat{\mathbf{d}}$ retains. Then*

$$\begin{aligned} \xi(x, u) &= a(x) + b(u), \\ a_i(x) &= \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u), \end{aligned} \tag{23}$$

576 *both taken modulo the all-ones vector. Consequently every reachable direction*
 577 *set is the same set translated: $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.*

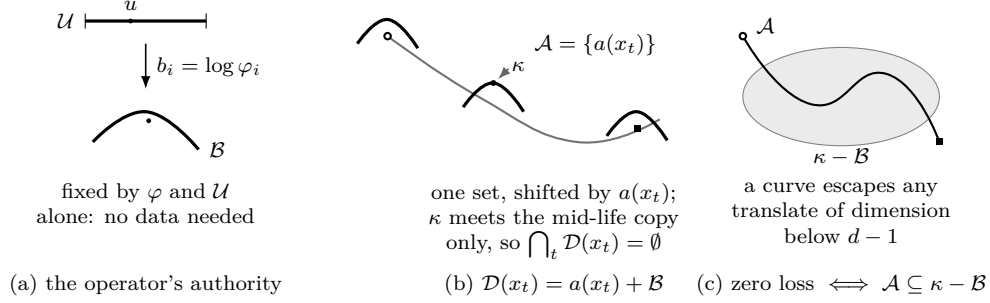


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the whole curve to lie in one translate (Proposition 4.4).

578 *Proof.* Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the
 579 product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along
 580 the all-ones vector. $\square$

581 The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the
 582 stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator's authority
 583 can therefore be computed offline from the accelerated-life model, with no
 584 degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage
 585 shapes h and the noise levels σ , and every reachable set is that one set carried
 586 along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the
 587 plant's demand $\mathcal{A}$ sticks out of the operator's authority $\mathcal{B}$.

588 That offline route is the most useful consequence drawn in this paper
 589 and also the most restrictive. It holds on the separable subclass and not
 590 beyond it, and the best high-fidelity turbofan data we have reject separability
 591 (Section 7). This subsection and Proposition 4.4 should therefore be read as
 592 describing a structured subclass that is convenient when it applies, not as
 593 a general account of real machinery. Fig. 6 marks the one branch of the
 594 development that depends on it.

595 **Proposition 4.4** (exact criterion). *Zero loss is attainable along a trajectory*
 596 *if and only if $\mathcal{A}$ is contained in a translate of $-\mathcal{B}$. In particular $\mathcal{A}$ must lie*
 597 *in an affine subspace of dimension at most $\dim \mathcal{B} \leq m$, which for a curve*
 598 *in the $(d-1)$ -dimensional quotient fails generically unless $m \geq d-1$; this*
 599 *recovers Theorem 4.1. When it does not fail, $m < d-1$ suffices.*

600 *Proof.* By Theorem 4.1, zero loss holds iff $q \in \mathcal{D}(x_t)$ for every t . Writing κ
601 for the quotient class of $\log \mathbf{v}$ and using Lemma 4.3, that reads $\kappa - a(x_t) \in \mathcal{B}$
602 for every t , that is $\mathcal{A} \subseteq \kappa - \mathcal{B}$. The dimension statement follows because a
603 translate of $\mathcal{B}$ spans an affine subspace of dimension $\dim \mathcal{B}$. $\square$

604 The containment is drawn in Fig. 3(c), and the mechanism the count
605 obscures is now visible. Alignment at an instant fixes u_t outright, since
606 $b(u_t) = \kappa - a(x_t)$ determines it; the input is spent, and any residual constraint
607 on the state must then hold along the flow with no freedom left to enforce
608 it. One input can carry the whole burden only when no residual constraint
609 is left, which happens when $\mathcal{A}$ already lies parallel to $\mathcal{B}$.

610 **Remark 4.5** (a family with $m = 1$ and any d). *Take exponential shapes*
611 *$h_i(x) = e^{\nu_i(x_i - x_i^0)}$, which (2) admits, and Arrhenius factors. Under the ramp*
612 *$u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 =$*
613 *$\varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = rE_i/\varphi_i^0$ makes*
614 *that velocity parallel to E , which is the direction of $\mathcal{B}$, and $b(u_t)$ retreats*
615 *along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Integrating the*
616 *construction for $d = 2$ to 8 with one input holds the misalignment below*
617 *1.5×10^{-6} degrees and the mean loss below 10^{-15} . These families carry no*
618 *prior variance of their own, so their losses are evaluated at a fixed $w = 0.9$*
619 *rather than at a γ the plant supplies; the bound has fifteen orders of magnitude*
620 *of headroom and does not turn on the choice. That is against 1.2×10^{-5} to*
621 *5.4×10^{-4} for the best constant input on the same plant.*

622 This bears on the sharpness of Theorem 4.1. The family of Remark 4.5 is
623 a knife edge. Perturbing the shape coefficients ν by 1% raises the mean loss
624 to 3.2×10^{-4} under the myopic policy, against 1.8×10^{-4} for the best constant
625 input on the same perturbed plant. At one percent of shape error, steering
626 is already worse than not steering. Closing the loop halves the damage but
627 does not repair it, so the fragility belongs to the construction and not to the
628 open-loop schedule. Mistuning the activation energies instead is tolerated to
629 about two percent. A designer should read Proposition 4.4 as an explanation
630 of when the counting rule binds, and Theorem 4.1 as the rule to design
631 against.

632 4.2. Invisibility to Fisher-information criteria

633 A natural objection is that all of this should follow from designing the in-
634 put to maximise the Fisher information about R , as in the observability-aware
635 literature discussed in Section 1. It does not, and the reason is structural.

636 **Proposition 4.6** (Fisher blindness). *The loss of a fixed indicator is not a*
637 *function of the Fisher information about R : configurations with equal Fisher*
638 *information can have different losses, and configurations with equal losses*
639 *can have different Fisher information. No criterion depending on the Fisher*
640 *information alone can therefore rank fixed indicators by the loss they incur.*

641 *Proof.* For a scalar latent the Fisher information is the number $\|\mathbf{d}\|^2 \text{Var}[R]$,
642 so trace, determinant and least eigenvalue coincide and every scalarisation is
643 a strictly monotone function of it. Under the rescaling $\mathbf{d} \mapsto c\mathbf{d}$ with $c > 0$,
644 that number changes while $\hat{\mathbf{d}}$, and hence β and ℓ , do not. A common gain
645 therefore equalises the scores of two configurations with different losses, and
646 separates the scores of two configurations with the same loss. $\square$

647 Stated in engineering terms, a Fisher criterion measures the strength of
648 the signal, whereas the loss of a fixed indicator depends on its direction.
649 Amplifying every channel by the same factor increases the first and leaves
650 the second unchanged. Section 6 exhibits this concretely: with the Fisher
651 score held equal to seven digits, the loss still spans a factor of 9.4.

652 5. The best achievable loss as a fractional control problem

653 Theorem 4.1 bounds the loss along a given trajectory. But the input
654 steers the trajectory as well as the misalignment on it, so a fundamental
655 limit requires optimising over both at once. This changes the structure of
656 the problem.

657 The objective is a time average over a lifetime that the input itself deter-
658 mines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (24)$$

659 Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes
660 (24) a fractional optimal control problem, meaning that the objective is a
661 ratio rather than that the dynamics carry a fractional-order derivative. The
662 fraction is not a technicality: it is the source of the trade-off, as follows.

663 Suppose T_f were fixed. Nothing in (2) penalises the input directly, so
664 the answer would be trivial: find whichever admissible u minimises β and
665 stay there. Because T_f is itself controlled, the denominator of (24) rules that
666 solution out. The operating point that aligns the indicator is in general not
667 the operating point that prolongs life, so aligning now shortens the interval

668 over which the alignment is averaged. The optimum must trade the two,
 669 and Section 6.5 measures how steep that trade is on a concrete plant. A
 670 fixed-horizon formulation represents none of it, which is why one was tried
 671 and discarded (Section 6.5).

672 5.1. A converse bound uniform over trajectories

673 Before computing anything we record a certificate that removes the con-
 674 ditioning. Theorem 4.1 bounds the loss along a trajectory the input has
 675 already chosen; the following proposition bounds it over all of them at once,
 676 and it is what makes the number computed below a limit rather than an
 677 observation.

678 **Proposition 5.1** (verification certificate). *Fix an indicator q and a constant*
 679 *$\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood*
 680 *of the admissible state region such that*

- 681 (i) $V(x) \leq 0$ for every x on the failure surface $\{\max_i x_i = 1\}$, and
- 682 (ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible
- 683 state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (25)$$

684 Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if
 685 $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence
 686 over all trajectories.

687 *Proof.* Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1
 688 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and
 689 the chain rule gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost
 690 every t . Applying (25) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e.,
 691 and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

692 the last step by (i), since x_{T_f} lies on the failure surface. Dividing by $T_f > 0$
 693 gives the claim. $\square$

694 Condition (25) constrains the plant data alone: it quantifies over every
 695 state and every admissible input, and mentions no trajectory. Any V one

696 can exhibit therefore certifies a limit, and the search for a converse becomes
 697 the search for a verification function.

698 The C^1 hypothesis cannot be weakened to Lipschitz continuity. A Lip-
 699 schitz V is differentiable almost everywhere in the state by Rademacher’s
 700 theorem, but a single trajectory is a null set and could in principle run inside
 701 the exceptional set, so the chain rule step would be unavailable; recovering
 702 the argument at that generality needs the Clarke generalised gradient in
 703 place of ∇V . Smoothness is also, as it turns out, exactly what the natural
 704 candidates lack.

705 We tried two. The value function of the Dinkelbach problem below sat-
 706 isfies (25) with equality in the viscosity sense [25], but value functions of
 707 this class are only Lipschitz. Computed on a 41^3 grid and tested on a 61^3
 708 one, (25) fails at about 45% of test points by up to 2.9×10^{-3} . Halving λ
 709 moves that violation by 0.6%, so the fault lies in the candidate and not in
 710 the bound. At the opposite extreme an affine $V(x) = c_0 - a_0^\top x$ has an exactly
 711 constant gradient and makes every requirement linear, so the best certifiable
 712 λ is a linear program; over 167,936 constraints it returns $V \equiv 0$ and only
 713 the pointwise bound $\min_{x,u} \ell$, which is vacuous here because the reachable
 714 floor falls to within sampling resolution of zero at mid-life. Deleting the two
 715 boundary requirements lets that program run to any cap imposed on it, which
 716 identifies them as what binds: trajectories of (2) increase in every coordinate,
 717 so an affine V is monotone along all of them at once and $V(x_0) \geq 0 \geq V|_{\text{failure}}$
 718 forces it to decrease, while the near-aligned states, where the running cost
 719 vanishes, demand a local increase.

720 The two failures are complementary, and together they say what a cer-
 721 tificate needs. The value function varies its gradient exactly as the two de-
 722 mands require but is not differentiable where that matters; affine functions
 723 are perfectly smooth but cannot vary their gradient at all. A certificate must
 724 therefore be both smooth and non-constant in its gradient, which points at
 725 a smooth under-approximation of the value function, obtained by sum-of-
 726 squares programming when the plant data are polynomial, or by interval
 727 arithmetic with an explicit Lipschitz constant. We do not claim a certified
 728 converse here, and the limit below remains a computation; what Proposi-
 729 tion 5.1 buys is the replacement of an open problem without shape by a
 730 concrete object with a checkable property.

731 Problems of the form (24) are handled by Dinkelbach’s transform [26]:

732 setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (26)$$

733 the map F is decreasing in λ and vanishes exactly at the optimal ratio, so λ^* is
 734 found by bisection on $F(\lambda) = 0$. The inner problem in (26) has free terminal
 735 time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman
 736 equation on the state alone, with no time argument. Because $\dot{x} > 0$ the
 737 state advances monotonically, and a semi-Lagrangian scheme [27] converges
 738 by value iteration. One implementation detail matters: the scheme must use
 739 a fixed spatial step rather than a fixed time step, because degradation speeds
 740 vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at
 741 one end of the envelope or the other.

742 6. Numerical study

743 Sections 3–5 are analytical, and three of their claims are stated sharply
 744 enough to be tested rather than illustrated: that the loss is invisible to a
 745 Fisher criterion, that the counting rule $m \geq d - 1$ binds in both direc-
 746 tions on a generic plant, and that the best achievable loss is a finite number
 747 one can compute. This section instantiates the model on a synthetic three-
 748 mechanism plant on which every quantity is available in closed form, checks
 749 those three claims against it, measures what steering is worth, and reads off
 750 the dimensionless index that says in advance whether it will repay the effort.
 751 Section 7 then reports what measured fleets can and cannot establish about
 752 the premises.

753 6.1. The test plant

754 We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in
 755 which the input u is the reciprocal temperature $1/(kT)$, so that this is the
 756 Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We
 757 set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with
 758 $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms.
 759 There is a single input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts
 760 zero loss out of reach unless this plant is one of the exceptions of Remark 4.5;
 761 Section 6.3 computes the floor and finds that it is not. Noise is unit and the
 762 prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of Lemma 2.2
 763 is evaluated at the plant’s own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than
 764 held fixed; γ varies along a trajectory because $\|\mathbf{d}\|$ does. Every trajectory

Table 2: Loss of the fixed indicator at a Fisher score held equal. The exposure factor δ , defined in the text, holds $\delta^2 \|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\delta^2 \ \mathbf{d}\ ^2$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	6.23	0.0215
1.0	0.628	0.4307565	4.53	0.0123
1.5	1.000	0.4307565	6.22	0.0208
2.0	1.564	0.4307565	7.53	0.0379
3.0	3.769	0.4307565	6.57	0.1161

765 starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of
766 (3). Raising x_f to unity leaves every qualitative conclusion below intact and
767 moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° ,
768 and the gain of Section 6.5 falls from 68 to 58. No constant reported below
769 should be read across thresholds.

770 *Verification of the identities.* Before using them we checked the analytical
771 results numerically. The rotation identity (20) holds to relative error below
772 1.1×10^{-7} over the five input-channel pairs $(d, m) = (2, 1), (3, 2), (3, 1),$
773 $(4, 2)$ and $(5, 3)$; the replicator form to 6.9×10^{-8} ; Theorem 3.6 to 5.1×10^{-6} ,
774 limited by finite differencing; and (17) to 1.1×10^{-16} .

775 6.2. A Fisher level set of varying loss

776 Proposition 4.6 is easy to demonstrate on this plant. For a constant input,
777 a common gain on $\mathbf{d}$ moves $\|\mathbf{d}\|$ while leaving $\hat{\mathbf{d}}(\tau)$, and hence β , untouched:
778 the Fisher score and the loss can be varied independently. Write $\delta(u)$ for the
779 gain that equalises the score across inputs, fixed by $\delta(u)^2 \|\mathbf{d}\|^2(u) = F$ with
780 F its value at $u = 1.5$. It is realised by rescaling how long the component
781 runs under load, which is why it does not rotate $\hat{\mathbf{d}}$. By construction $\delta = 1$
782 at $u = 1.5$, and $\delta > 1$ wherever the raw score is smaller, so it is a multiplier
783 on exposure rather than a fraction of it. Table 2 does exactly that, holding
784 the Fisher score equal to seven digits by tuning the exposure factor δ . The
785 loss still spans a factor 9.4.

786 The two objectives also disagree on which input to prefer: the raw score,
787 before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and
788 the loss is 1.7 times its value at $u = 1$. How far they disagree is a property
789 of the instance; what is structural is only the blindness.

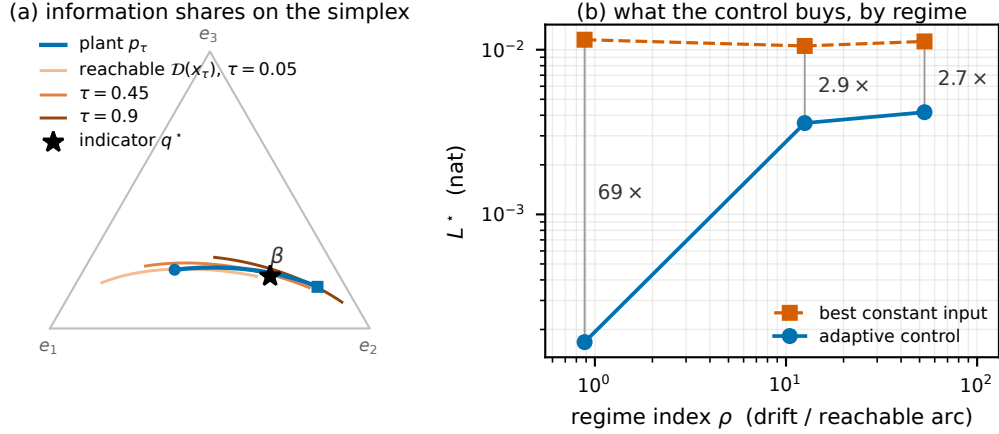


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* . No point lies on all three, so β cannot be held at zero (Theorem 4.1). (b) What steering buys, against the regime index ρ .

6.3. The reachability floor with a single input

With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} = 4.52^\circ$; it falls through mid-life to below the 0.01° resolution of our sampling, where the indicator is momentarily reachable, and rises again towards failure (Fig. 5(a)). Zero average loss would need the indicator reachable at every instant and not at one, so it is unattainable however the input is steered, in agreement with Theorem 4.1. Fig. 4(a) shows the geometry: the reachable set sweeps across q^* as the plant ages, so the mid-life arc passes close to the indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

6.4. The counting rule, tested in both directions

Theorem 4.1 is an equivalence, and the paragraph above tests only half of it: with too few inputs the floor stays positive. The other half, that supplying the inputs the counting rule asks for makes the floor vanish, is the more useful claim for a practitioner deciding how many operating variables to instrument, so we test it directly.

Give the same three-mechanism plant a second independent input by writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m = 2 = d - 1$ and each input acts on the mechanisms in a different pattern.

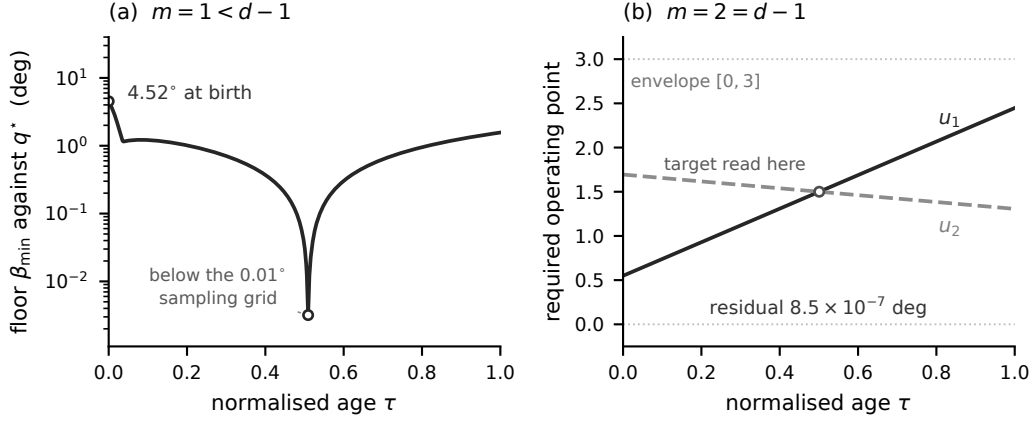


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly; plotted is the operating point this requires, which never leaves $\mathcal{U}$.

809 Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction fixes the
 810 $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns,
 811 with a coefficient matrix of determinant 0.27 here, hence a unique solution
 812 at every state.

813 Taking as target the direction the plant occupies at mid-life under a
 814 central operating point, that solution lies inside the admissible box $[0, 3]^2$
 815 at every one of 600 sampled states covering the whole life, and the residual
 816 misalignment there is 8.5×10^{-7} degrees, that is, zero to numerical precision;
 817 the schedule it requires is drawn in Fig. 5(b). An independent 90×90 grid
 818 search over the box returns a floor of at most 0.156° , which is the resolution
 819 of the grid rather than a property of the plant.

820 The contrast with the single-input case is the content of the theorem: the
 821 same mechanisms, the same shapes, the same envelope on each input, and the
 822 peak floor moves from 4.52° to zero at every sampled state on the addition
 823 of one operating variable. On this plant the counting rule is therefore not
 824 a conservative sufficient condition but a sharp one, and within the adopted
 825 model it fixes how many independent control degrees of freedom the plant
 826 must offer. How those degrees of freedom correspond to physical actuators,
 827 and how the mechanisms are individuated in the first place, are modelling
 828 questions this paper does not settle. Proposition 4.4 says which plants escape
 829 it, and Remark 4.5 says why a designer should not count on being one of
 830 them.

831 6.5. *The value of steering*

832 Minimising over q gives $q^* = (0.466, 0.771, 0.435)$, at which the adaptive
833 policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input,
834 a factor of 68. The ratio depends on how finely $\mathcal{U}$ is discretised for the policy,
835 rising to 72 at 161 points; we report it on the 41-point grid used throughout.

836 The factor is a ratio of information losses, not of information, and it is
837 not free in reliability terms: the policy that attains it runs the component
838 differently, and Section 6.6 finds the two closed loops separated by a fac-
839 tor of 3.4 in life. Aligning the indicator and prolonging the component are
840 competing objectives, which is what makes the horizon in (24) a controlled
841 quantity rather than a constant. That factor is the value of steering at all,
842 not the value of steering optimally: a myopic policy that attends only to the
843 numerator of (24) sits 2.8% above the optimum at q^* .

844 The fraction in (24) is doing visible work at this operating point. The
845 closed loop at q^* and the closed loop at a balanced indicator, which calls for
846 far less steering, differ by a factor 3.4 in lifetime: what alignment buys in the
847 numerator is paid for in the denominator. This is the tension anticipated in
848 Section 5, and it is why the horizon cannot be fixed in advance.

849 6.6. *The fractional limit*

850 Dinkelbach bisection with the control grid matched to the policy gives,
851 over $N \in \{25, 29, 35\}$, extrapolants agreeing to 1.2% and $\lambda_\infty^* = 1.63 \times 10^{-4}$,
852 below the realisable 1.68×10^{-4} as it must be.

853 We quote this to two digits only, for the following reason: the raw values
854 are twice the extrapolant, so the first-order correction carries half the answer,
855 and both quantities still move by 4% between control grids of 41 and 81
856 points. Three checks guard the number: F must decrease and cross zero, λ^*
857 must not exceed a realisable policy, and the grid sequence must extrapolate
858 consistently. A fixed-horizon formulation failed the second and third and was
859 discarded, as did a first attempt with mismatched control grids. A converse
860 bound uniform over trajectories, which would replace this computation by a
861 theorem, remains open.

862 6.7. *The regime index, and how far it screens*

863 For a practitioner the first question is not how to steer but whether to
864 bother. One dimensionless ratio answers it. The reachable set $\mathcal{D}(x)$ is an
865 arc; write ρ for the drift of its midpoint over the life divided by its mean

Table 3: The regime index. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under adaptive control, and the gain its ratio to the best constant input.

E	arc	ρ	L^*	gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.7×10^{-4}	68×
(0.80, 0.82, 0.79)	1.93°	12.5	3.6×10^{-3}	2.9×
(0.80, 0.805, 0.798)	0.45°	53.1	4.2×10^{-3}	2.7×

length, so ρ measures how far the informative direction wanders in units of the actuation authority available to chase it.

Table 3 varies the spread of the activation energies, which by (20) is what sets that authority. As the arc collapses, the attainable loss L^* rises by a factor 25 and the value of control falls from 68 to 2.7: with weak authority there is little left to steer. Fig. 4(b) plots the two against ρ . The construction of this paper is a statement about the low- ρ regime, and ρ can be computed from a nominal degradation model before any indicator is designed.

Three plants differing only in their activation energies are not a screening law, so the index was tested on 299 further plants drawn at random: two to five mechanisms, one or two independently adjustable inputs, and the pre-exponentials, activation energies, shape coefficients, input envelope, initial state and signal-to-noise ratio varied together. The rank correlation between ρ and the gain is -0.36 , and it keeps its sign inside every stratum of the mechanism count and of the input count, so it is not an artefact of either.

Its useful direction is the optimistic one. Of the plants with $\rho < 1$, 53% repay steering by a factor of five or more, against 32% over the whole sample. As a veto it is weaker: rejecting every plant with $\rho \geq 1$ would also reject 23% of plants that would in fact have repaid a factor of five, and that is the expensive error rather than the cheap one. Most of the index’s discriminating power is in its denominator: the reachable arc alone ranks the plants almost as well, at 0.35 in magnitude against the ratio’s 0.36, while the drift on its own ranks them at below 0.05 in magnitude, which is nothing. The drift earns its place only when plants of equal arc are compared. The first row of Table 3 is also not typical of its own regime: among sampled plants within a factor 1.4 of its ρ , the median gain is 5.2 rather than 68. ρ indicates a regime; it does not decide a design.

893 6.8. What the loss buys a downstream estimator

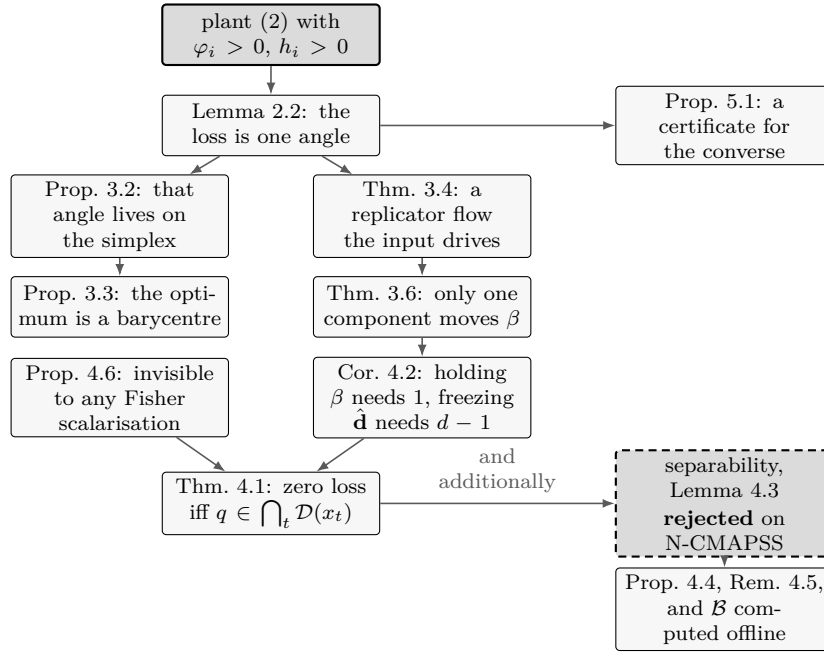
894 Corollary 2.4 prices the loss in the error of the estimate the model itself
 895 prescribes. A practitioner does not use that estimate, so the ordering is worth
 896 checking against a procedure that uses none of the construction: sample the
 897 indicator over the last 40% of life, match the observed history against the
 898 indicator’s own reference curve, and read remaining life off the best-fitting
 899 time offset. That estimator sees a scalar signal and nothing else — not $\hat{\mathbf{d}}$,
 900 not γ , not the loss.

901 Over twelve indicators spanning $\bar{\ell}$ from 0.012 to 0.37, its median absolute
 902 error — one of the standard prognostic accuracy measures [28] — rises with
 903 the loss, at Spearman correlation 0.89, and the best-aligned indicator beats
 904 the worst by a factor 1.5. The factor is far smaller than the 31-fold spread in
 905 $\bar{\ell}$, and it should be: the loss is a logarithmic information quantity, the error
 906 is a time, and an estimator with the whole history recovers much of what
 907 any one instant discards.

908 7. What public data can say about the premises

909 The construction rests on three modelling choices, and each is empirically
 910 testable. They do not carry equal weight, and Fig. 6 separates what depends
 911 on which. Two are premises about the plant: that the informative direction
 912 rotates along the trajectory, and that the input moves it. The third is the
 913 separability of Lemma 4.3, which is what makes the operator’s authority
 914 computable offline. Table 4 summarises what six public run-to-failure fleets
 915 currently settle. The protocols are not reproduced here; the datasets are
 916 named so that each test can be located and repeated.

917 Three readings qualify the table. The Severson channels are capacity fade,
 918 resistance growth and mean temperature, and mean temperature is a moni-
 919 toring channel rather than a damage state — it is the operating variable itself
 920 in the Arrhenius form of Section 2. That row therefore reads the rotation
 921 of a direction formed from three measured series, which is what the premise
 922 asserts, and does not need (5)’s pairing of channels with mechanisms. On the
 923 four measured fleets the obstacle to testing actuation is the estimator and not
 924 the physics, so the premise is confirmed in simulation and open on hardware.
 925 And the C-MAPSS contrast is reported against a negative control — fake
 926 regimes cut from a single real one, where no contrast exists by construction
 927 — because a procedure that returned a positive contrast regardless would
 928 produce the same table entry.



Every result on the spine needs only the positivity of the degradation model. Only the dashed branch needs separability, and it is that hypothesis, not the geometry, that Section 7 reports as rejected on the best public turbofan data.

Figure 6: What rests on what. The spine needs only the positivity of the degradation model. Only the dashed branch adds separability, which Section 7 reports as rejected on the best public turbofan data.

929 The consequence is the scope of the results above. The geometry of
 930 Sections 3 and 4 needs only $\varphi_i > 0$ and $h_i > 0$, and is untouched. What
 931 separability buys, and what the rejection withdraws, is Proposition 4.4 and
 932 the offline route to $\mathcal{B}$: those describe a model class, and a plant must be
 933 shown to belong to it before they are used. The two-way layout above is the
 934 test, and it costs one pass over data a reliability programme already holds.

935 8. Discussion

936 Four practical readings follow from the results above. The first two hold
 937 whatever the plant is, because one is a test and the other an invariance. The
 938 last two assume the separable form of (2), which Section 7 reports as rejected
 939 on the only data able to test it, so they apply where that form does and not
 940 otherwise.

Table 4: What the six public fleets settle, and what they cannot. Only the two turbofan simulations vary the operating point within a unit, which is why the actuation premise is confirmed in simulation and open on measured hardware.

fleet	premise	what is measured	reading
Severson lithium-ion [29]	rotation	arc of the fleet-mean informative direction over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); stays in the open positive orthant once the window reaches 301 cycles
XJTU-SY [30], PRONOS-TIA [31]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [32]	actuation	—	four cells
C-MAPSS [33]	actuation	between-regime minus within-regime angle, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; fake regimes give $ B - W \leq 0.5^\circ$
N-CMAPSS DS02, DS01 [34]	separability	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected

941 Test the separability assumption before relying on it. The two-way layout
942 of Section 7 needs only a fleet whose operating point varies within a unit,
943 and costs one pass over data a reliability programme already holds. If the
944 interaction clears the measured noise floor, the offline route to the operator’s
945 authority is unavailable and the two model-based readings below do not
946 apply.

947 No Fisher criterion can rank fixed indicators by the loss they incur. This
948 is the second of the two unconditional readings: Proposition 4.6 is an in-
949 variance of the loss under a common gain and holds whatever the plant, so
950 rules that maximise a scalarisation of the Fisher information cannot separate
951 configurations whose losses differ by an order of magnitude. Where the two
952 objectives do coincide, that is a property of the application, to be verified
953 rather than assumed.

954 The binding constraint is actuation authority, not indicator sophistica-
955 tion. Theorem 4.1 prices the operating variables a lossless fixed indicator
956 would need, and Proposition 4.4 sharpens the count to sufficient but not

necessary, with exceptions that Remark 4.5 shows to be a knife edge. Care in choosing the weights substitutes for actuation in neither case. What one controllable variable does buy is the weaker objective of Corollary 4.2: an installation that could never freeze the informative direction can still hold the misalignment constant, though not at zero. Authority bought this way is not free, and Section 6.6 prices it in life.

The decision to steer precedes the design of the indicator. The regime index ρ of Section 6.7 requires only a nominal degradation model and the operating envelope, and Table 3 shows the value of steering collapsing from $68\times$ to under $3\times$ as the reachable arc shortens relative to the drift. It indicates a regime rather than deciding a design: Section 6.7 measures how often it is wrong, and in which direction.

Limitations. Four limitations qualify the readings above. The separable form of (2) is not only an idealisation but one the data reject: Section 7 finds the interaction between operating point and state larger than either main effect on two N-CMAPSS datasets, so computing the operator’s authority offline is available only where that form holds. The fundamental limit of Section 5 is computed rather than proved, to two digits. The actuation premise is confirmed on simulated rather than measured data, for the reasons set out in Section 7. And the population is taken to slide along one degradation path, which is what makes remaining life a coordinate on it; under dispersion in the coefficients of (2) the sensitivity that Remark 2.1 measures stands some 45° from (6), so the construction describes fleets that vary in when they fail rather than in how they degrade.

9. Conclusion

A fixed health indicator discards information, and this paper has treated what it discards as a quantity the operating point controls. Indicator design then becomes a reachability question. The family of reachable direction sets bounds what the operating point can achieve, and their drift relative to their width is a dimensionless measure, available before any indicator is fixed, of whether substantial benefit is plausible. Actuation authority, rather than sophistication of the indicator, governs whether a fixed compression can come close to lossless. Whether that authority is present is an empirical question rather than a modelling assumption, and on 439 simulated turbofan records it is. Establishing the same on measured fleets awaits records in which the operating point varies enough to show the effect.

993 Section 7 rejects that separable form on the highest-fidelity public tur-
994 bofan data, so the geometry of Section 4.1, and the offline route it opens,
995 describe a model class rather than measured machinery. What replaces that
996 structure when the stress model is not separable, and whether a weaker one
997 still yields a computable reachability criterion, remains open.

998 Two further questions remain open, the first of them now in a more
999 specific form. Proposition 5.1 reduces the search for a converse bound uni-
1000 form over trajectories to the exhibition of a single continuously differentiable
1001 verification function; the value function is the natural candidate and fails
1002 the differential test for a structural reason, so what is required is a smooth
1003 under-approximation of it, by sum-of-squares programming or by interval
1004 arithmetic with an explicit Lipschitz constant. The other is unchanged: the
1005 fractional limit to better than two digits, which requires a scheme of higher
1006 order than the semi-Lagrangian one used here.

1007 **Data availability and reproducibility**

1008 The six fleets named in Section 7 are publicly available. One script re-
1009 generates every computed result from a single configuration, and a com-
1010 panion check recomputes each reported number and compares it with the
1011 value written in the text, covering the tables and the quantities reported
1012 alongside them. Those checks are maintained against the full manuscript
1013 from which this paper is drawn, and all of them pass; the numbers re-
1014 ported here are carried over from it unchanged. Three groups of num-
1015 bers lie outside it: the residuals of the identity checks in Section 6, the
1016 barycentre comparison in Section 3.3, and the diagnostics of the two dis-
1017 carded formulations in Section 5. The plots are produced by three further
1018 scripts. All of these are available at [https://github.com/HuyHoangLe0201/](https://github.com/HuyHoangLe0201/health-indicator-information-loss)
1019 [health-indicator-information-loss](https://github.com/HuyHoangLe0201/health-indicator-information-loss).

1020 **Declaration of competing interest**

1021 The authors declare that they have no known competing financial inter-
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